modulations (larger amplitude here, smaller there; higher frequency wiggles here, lower there), so the coalescence history is encoded in modulations of the curvature ripples. And just as the sound waves carry their encoded symphony from the orchestra that produces it to the audience, so the curvature ripples carry their encoded history from the coalescing holes to the distant Universe.

The curvature ripples travel outward in the fabric of spacetime, through the agglomerate of stars and gas where the two holes were born. The agglomerate absorbs none of the ripples and distorts them not at all; the ripples' encoded history remains perfectly unchanged. On outward the ripples propagate, through the agglomerate's parent galaxy and into intergalactic space, through the cluster of galaxies in which the parent galaxy resides, then onward through one cluster of galaxies after another and into our own cluster, into our own Milky Way galaxy, and into our solar system, through the Earth, and on out toward other, distant galaxies.

If we humans are clever enough, we should be able to monitor the ripples of spacetime curvature as they pass. Our computers can translate them from ripples of curvature to ripples of sound, and we then will hear the holes' symphony: a symphony that gradually rises in pitch and intensity as the holes spiral together, then gyrates in a wild way as they coalesce into one, deformed hole, then slowly fades with steady pitch as the hole's protrusions gradually shrink and disappear.

If we can decipher it, the ripples' symphony will contain a wealth of information:

1. The symphony will contain a signature that says, "I come from a pair of black holes that are spiraling together and coalescing." This will be the kind of absolutely unequivocal black-hole signature that astronomers thus far have searched for in vain using light and X-rays (Chapter 8) and radio waves (Chapter 9). Because the light, X-rays, and radio waves are produced far outside a hole's horizon, and because they are emitted by a type of material (hot, high-speed electrons) that is completely different from that of which the hole is made (pure spacetime curvature), and because they can be strongly distorted by propagating through intervening matter, they can bring us but little information about the hole, and no definitive signature. The ripples of curvature (gravitational waves), by contrast, are